Syed Ibrahim Omer

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Summary: Undergraduate researcher in Data Science working on state space models and efficient ML. First author of MaRK (NeurIPS 2026), a conditioning and fine-tuning framework that adapts state space models to diffusion objectives, reducing trainable parameters by 90% while establishing stability and operator recovery guarantees. Research interests center on function-space analysis of sequence models: studying operators by their action on inputs rather than parameter geometry, so that architectural modifications carry provable behavioral guarantees.

Education

City University of Hong Kong Expected May 2028

Bachelor of Science (BS) in Data Science

Distinctions: Dean's List, Full Academic Scholarship, Tiger Programme Member, Diversity Grant Recipient

Relevant Coursework: Linear Algebra, Probability & Statistics, Advanced Statistics, Bayesian Analysis, Convex Optimization, Multi-Variable Calculus

Experience

NVIDIA AI Technology Center (NVAITC) Hong Kong SAR

Student Ambassador (Research Track) *June 2025 – Present*

- First author of **MaRK** (accepted to NeurIPS 2026), contributed to the methodology, source code, proofs, and rebuttal
- Co-authoring **BXINT4** with NVAITC researchers, a method for 4-bit training on older NVIDIA GPUs (Turing to Lovelace) and edge devices, including custom CUDA kernels and performance profiling

Habib Bank Limited (HBL) Karachi, Pakistan

Machine Learning Intern, Data Lake Function *Jun 2026 – Aug 2026*

- Built account-level models that predict which customers will churn and when, replacing branch-level reports that could only describe churn after it happened
- Engineered scalable ETL pipelines in DuckDB to transform raw data lake records into statistical tables, used Microsoft Research's **LightGBM** engine for memory-efficient training
- Achieved **0.93 weighted F1** (0.67 macro F1) across evaluation accounts. Delivered churn hazard risk curves, what-if analysis pipelines for peak risk timing, and **TreeSHAP** feature attributions, presenting actionable insights to non-technical business stakeholders

Publications

MaRK: Markov-adapted Recurrent Kernels | *First Author* NeurIPS 2026

- Designed MaRK (Markov-adapted Recurrent Kernels), a conditioning framework that directly modulates recurrent operators tailored to the diffusion context
- **Reduced trainable parameters by 90%** to transition a frozen 111M-parameter Hydra backbone from a bidirectional to iterative diffusion objective by designing 3 novel adapter geometries with 7.5–13M parameters (Hypernet, Chebyshev, and DCT)
- Established asymptotic stability across all model layers by mathematical proof and empirically verifying positive stability bounds ($\varepsilon > 0$) under extreme modulation
- **Derived explicit $O(1/\sqrt{N})$ operator recovery convergence bounds** by proving Rademacher complexity bounds for dependent sequences, validating the theoretical trend via synthetic kernel experiments across 15 seed runs
- Built training pipeline on NVIDIA NeMo with multi-GPU DDP on 2× GH200s; formally verified proofs in Lean

Redefining Hybrid Blockchains: A Balanced Architecture | *Sole Author* Pre-print

- Designed a blockchain architecture that reduces network traffic from $\mathcal{O}(N^2)$ to $\mathcal{O}(N)$, reaching **1,030 transactions per second** and **2.2 seconds** median time to finality
- Independently cited in peer-reviewed literature; open-sourced the full implementation (MIT License) on GitHub

Skills

Programming Languages: Python, C/C++, Rust, SQL, Lean 4
Software/Frameworks: NeMo, CUDA/CUTLASS, Nsight Compute/Systems, HuggingFace, Transformers, Datasets, PyTorch, Docker, Kubernetes, Apache Spark, Apache Kafka
Languages: English (IELTS Score: 8.0), Urdu